

PART A

Circuit (a)

1. Total Resistance (R)

$$R1 + R2 + R3 = 20 + 30 + 50 = 100 \text{ ohms}$$

2. Total Current (I)

$$I = \text{Voltage} / \text{Total resistance} =$$

$$125\text{V} / 100 \text{ ohms} = 1.25 \text{ Amperes}$$

3. Current through each

$$\text{In a series, } I = I1 = I2 = I3 = 1.25 \text{ Amperes}$$

4. Voltage Drop (V)

$$V = I * R$$

$$V1 = 1.25 * 20 = 25 \text{ Volts}$$

$$V2 = 1.25 * 30 = 37.5 \text{ Volts}$$

$$V3 = 1.25 * 50 = 62.5 \text{ Volts}$$

5. Power Dissipated (P)

$$P = V * I$$

$$P1 = 25 * 1.25 = 31.25 \text{ Watts}$$

$$P2 = 37.5 * 1.25 = 46.875 \text{ Watts}$$

$$P3 = 62.5 * 1.25 = 78.125 \text{ Watts}$$

Circuit (b)

1. Total Resistance (R)

$$(1/R1) + (1/R2) + (1/R3) = 1/20 + 1/100 + 1/50 = 12.5 \text{ ohms}$$

2. Total Current (I)

$$I = \text{Voltage} / \text{Total resistance} =$$

$$125\text{V} / 12.5 \text{ ohms} = 10 \text{ Amperes}$$

3. Current through each

The voltage is the same across all

$$I1 = 125/20 = 6.25 \text{ Amperes}$$

$$I2 = 125/100 = 1.25 \text{ Amperes}$$

$$I3 = 125/50 = 2.5 \text{ Amperes}$$

4. Voltage Drop (V)

All are 125V because they are parallel

5. Power Dissipated (P)

$$P = V * I$$

$$P1 = 125 * 6.25 = 781.25 \text{ Watts}$$

$$P2 = 125 * 1.25 = 156.25 \text{ Watts}$$

$$P3 = 125 * 2.5 = 312.5 \text{ Watts}$$